16/04/2024

Phase-II
CODE-D

Aakash
Medical | IIT-JEE | Foundations

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FINAL TEST SERIES for NEET-2024

Test -5

MM : 720 **Time : 3 Hrs. 20 Mins.**

CC-026 CC-026 CC-026 CC-026

Topics covered :

Physics : Electromagnetic Induction, Alternating Current, Electromagnetic Waves, Ray Optics and Optical Instruments, Wave Optics

Chemistry : The *p*-Block Elements (Group 15-18), The *d* and *f*-Block Elements, Coordination Compounds, Haloalkanes and Haloarenes

Botany : Principles of Inheritance and Variation, Molecular Basis of Inheritance, Microbes in Human Welfare

Zoology : Human Reproduction, Reproductive Health, Evolution

Instructions :

(i) There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.

(ii) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.

(iii) Use blue/black ballpoint pen only to darken the appropriate circle.

(iv) Mark should be dark and completely fill the circle.

(v) Dark only one circle for each entry.

(vi) Dark the circle in the space provided only.

(vii) Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

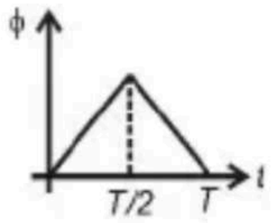
PHYSICS

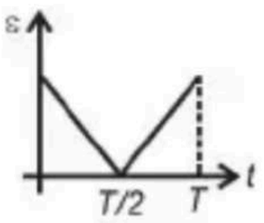
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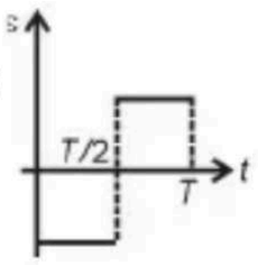
Choose the correct answer :

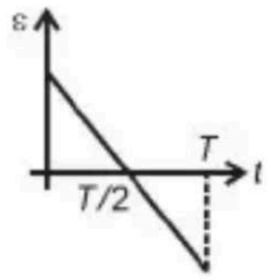
SECTION - A

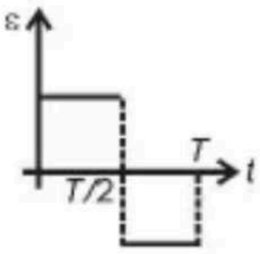
1. Magnetic flux linked with a coil changes with time as shown below its corresponding emf (ε) vs time (t) graph will look like



(1) 

(2) 

(3) 

(4) 

2. Two coils are inductively coupled in such a way that rate of change of current of 2 A/s in one coil induces an e.m.f. of 2 V in the neighbouring coil. The mutual inductance of coils is

(1) 1 henry (2) 0.5 henry

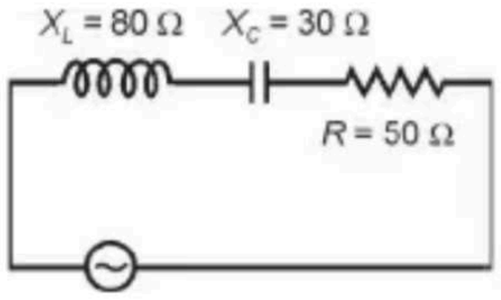
(3) 4 henry (4) e^4 henry

3. A long coil having 400 turns per meter carries a current of 0.5 A. The self inductance per unit volume in SI unit of this air cored solenoid is nearly
(1) 201×10^{-3} (2) 350×10^{-3}
(3) 70×10^{-3} (4) 100×10^{-3}

4. An alternating current varies as $I(t) = 20\sin(100\pi t)$. Its rms value in half cycle will be
(1) $\frac{10}{\sqrt{2}} A$ (2) $20A$
(3) $10\sqrt{2}A$ (4) Zero

5. Reciprocal of impedance of an A.C. circuit is known as admittance. The SI unit of admittance is
(1) ohm (2) mho
(3) ohm metre (4) mho per metre

6. Consider the following LCR circuit connected to an A.C. source $E = 200\sin(100\pi t)$. The instantaneous current as a function of time can be given as



(1) $2\sqrt{2}\sin\left(100\pi t - \frac{\pi}{4}\right)$
(2) $2\sin\left(100\pi t + \frac{\pi}{4}\right)$
(3) $4\sin(100\pi t)$
(4) $\frac{2}{\sqrt{2}}\sin\left(100\pi t - \frac{\pi}{4}\right)$

7. Which of the following is responsible for loss in a transformer?
(1) Eddy currents
(2) Inefficient coupling of coils
(3) Resistance of windings
(4) All of the above

8. An EM wave which is widely used in disinfecting and water purifiers is
(1) UV rays (2) Infra-red rays
(3) X-rays (4) γ-Rays

9. The variation of electric field in an EM wave is given by $E = 120\sin(3 \times 10^4 t - 2 \times 10^{-4} x)$. The amplitude of magnetic field is (All quantities are in SI units)
(1) 4×10^{-7} T (2) 8×10^{-7} T
(3) 40×10^{-7} T (4) 80×10^{-7} T

10. Consider the following statements and choose the correct option
(I) For a triangular prism, the sum of angle of prism and angle of minimum deviation is equal to the sum of angle of emergence and incidence.
(II) A graph between angle of incidence (at abscissa) and angle of deviation (at ordinate) for a triangular prism is parabolic in nature.
(1) Only (I) is correct
(2) Only (II) is correct
(3) Both (I) and (II) are correct
(4) Both (I) and (II) are incorrect

11. A compound microscope is of magnifying power 80. The magnifying power of its objective lens is 16. The magnification of its eyepiece lens will be
(1) 5 (2) 4
(3) 8 (4) 10

12. An object is kept at a distance of 10 cm in front of a concave lens of focal length 20 cm. The distance of image from the lens will be
(1) 10 cm on the same side of object
(2) 20 cm on the opposite side of object
(3) $\frac{20}{3}$ cm on the same side of object
(4) $\frac{10}{3}$ cm on the opposite side of object

Space for Rough Work

13. An equiconvex lens of refractive index 1.5 and radius 20 cm. The focal length of the lens is

(1) 10 cm (2) 40 cm

(3) $\frac{40}{3}$ cm (4) 20 cm

14. The path difference between two wavefronts emitted by coherent sources of wavelength 6280 Å is 2 μm. The phase difference between the wavefronts at that point is

(1) 2 radian (2) 20 radian

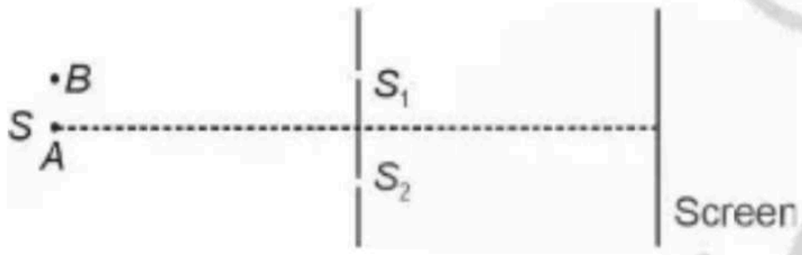
(3) 0.5 radian (4) 5 radian

15. A beam of light consisting of two wavelength 6300 Å and λ Å is used to obtain interference fringes in a Young's double slit experiment. If 4th bright fringe of 6300 Å coincides with 5th dark fringe of λ Å, the value of λ (in Å) is

(1) 5200 (2) 4800

(3) 5600 (4) 6200

16. In YDSE, the source placed symmetrically with respect to the slit is now moved parallel to the plane of the slits from point *A* to point *B* so that it is closer to the upper slit, as shown, Then



(1) The fringe width will increase, and fringe pattern will shift down

(2) The fringe width will remain same, but fringe pattern will shift up

(3) The fringe width will decrease, and fringe pattern will shift down

(4) The fringe width will remain same, but fringe pattern will shift down

17. The magnetic flux through a coil is varying according to the relation $\phi = (5t^3 + 4t^2 + 2t - 5)$ Wb. Calculate the induced current through the coil at $t = 2$ s, if the resistance of the coil is 5 Ω.

(1) 7.8 A (2) 15.6 A

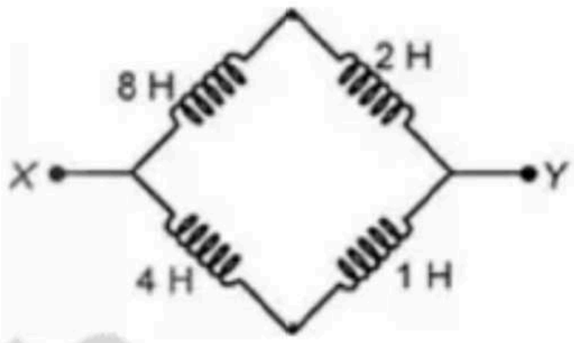
(3) 32.33 A (4) 61 A

18. What inductance would be needed to store 1 kWh of energy in a coil carrying a 200 A current?

(1) 90 H (2) 180 H

(3) 45 H (4) 360 H

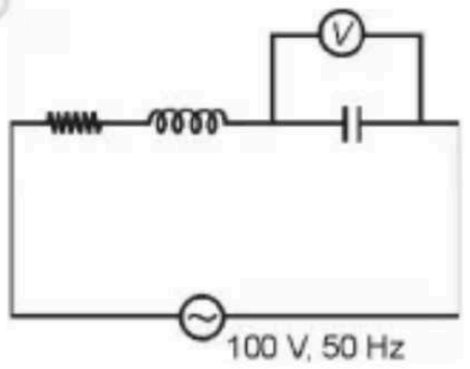
19. In the given network, the equivalent inductance between *X* and *Y* will be approximately (neglect mutual inductance in circuit)



(1) 15 H (2) 3.3 H

(3) 4.4 H (4) 1.1 H

20. Reading of voltmeter *V* shown in figure at resonance is 400 V, then the quality factor of the circuit is



(1) 2

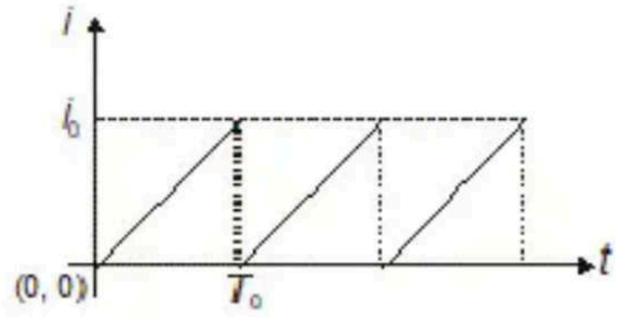
(2) 4

(3) 1

(4) 3

Space for Rough Work

21. Current (i) in a circuit varies with time (t) as shown



The rms value of current in time interval zero to T_0 is

(1) i_0 (2) $\frac{i_0}{\sqrt{3}}$

(3) $\sqrt{\frac{2}{3}}i_0$ (4) $\sqrt{\frac{3}{2}}i_0$

22. A transformer

(1) Convert AC into DC

(2) Convert DC into AC

(3) Is employed to obtain a suitable AC voltage

(4) Is employed to obtain a suitable DC voltage

23. In an AC circuit, voltage leads the current by 30°. Then circuit can be

(1) Only LCR (2) Only LR

(3) Only CR (4) Both LCR or LR

24. Amplitude of magnetic field in a beam of electromagnetic wave with intensity 3 Wm^{-2} approximately is

(1) 1.6×10^{-9} T (2) 1.6×10^{-7} T

(3) 1.6×10^{-8} T (4) 3.2×10^{-14} T

25. Two beams of light of intensities I_1 and I_2 interfere to give an interference pattern. If the ratio of maximum intensity to that of minimum intensity is $\frac{25}{9}$, then $\frac{I_1}{I_2}$ is

(1) $\frac{5}{3}$ (2) $\frac{7}{3}$

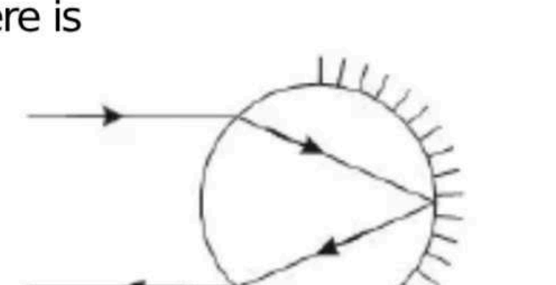
(3) $\frac{16}{1}$ (4) $\frac{8}{3}$

26. A beam of light of wavelength 600 nm from a distant source falls on a single slit, 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance between first dark fringes on either side of the central bright fringe is

(1) 1.2 cm (2) 1.2 mm

(3) 2.4 cm (4) 2.4 mm

27. A thin lens is broken into four equal parts and rearranged as shown. If the initial focal length was f, then after rearrangement the equivalent focal length will be



(1) $\frac{f}{4}$ (2) $\frac{f}{2}$

(3) $2f$ (4) $4f$

28. A fish inside water, looks upwards towards an unobstructed overcast sky. What total angle does the sky appear to subtend on fish? (Take refractive index of water as $\sqrt{2}$)

(1) 180° (2) 90°

(3) 75° (4) 60°

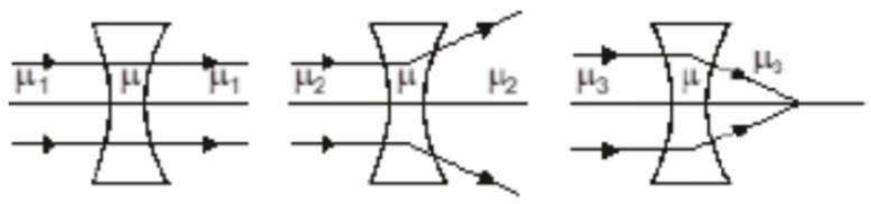
29. A transparent sphere kept in air has its right half polished so as to act as a mirror. A paraxial ray incident from left, exits parallel to the incident ray as shown. The refractive index μ of the material of the sphere is

(1) 1.2 (2) 1.5

(3) 1.8 (4) 2.0

Space for Rough Work

30. What is the relation between the refractive indices μ_2, μ_1 and μ_3, if the behaviour of light rays is as shown in figure?



(1) $\mu_1 = \mu_2 = \mu_3$ (2) $\mu_1 > \mu_3 > \mu_2$

(3) $\mu_3 > \mu_2 > \mu_1$ (4) $\mu_3 > \mu_1 > \mu_2$

31. Which of the following has got dimensions of time

(1) $\frac{R}{C}$ (2) $\frac{L}{R}$

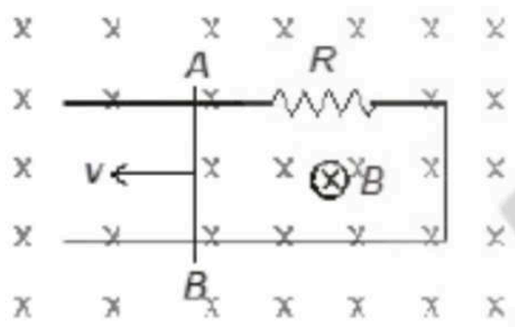
(3) $\frac{1}{\sqrt{LC}}$ (4) Both (1) and (2)

32. The reading of hot wire ammeter in sinusoidal a.c. circuit is 10 A. The peak value of current (approximately) is

(1) 7 A (2) 10 A

(3) 14 A (4) Insufficient data

33. As shown in figure a uniform magnetic field B exist in large space and a conducting rod *AB* is given a velocity *v* to the left. No friction is present between rails and rod then



(1) Rod will move with constant velocity

(2) Velocity of rod will decrease with time and will finally become zero

(3) No current will be induced in the circuit

(4) Direction of induced current will be clockwise

34. Read the following statements and choose the correct option.

Statement A: An emf can be induced by moving a conductor in a magnetic field.

Statement B: An emf can be induced in a coil by changing the magnetic field, in which the coil is lying.

(1) Both statements A and B are true

(2) Statements A is true but statements B is false

(3) Statements B is true but statements A is false

(4) Both statements A and B are false

35. Which of the following quantity remain constant when a light ray enters from air to glass?

(1) Wavelength (2) Frequency

(3) Velocity (4) Both (1) & (3)

SECTION - B

36. Column I contains mathematical operation of few quantities while column II contains the corresponding proportional results. Match the columns by choosing the correct option (symbols have usual meanings)

	Column I		Column II
A.	$\frac{d\phi_B}{dt}$	P	Magnetic flux
B.	$\int \vec{B} \cdot \vec{ds}$	Q	Induced electric field
C.	$\frac{d\vec{B}}{dt}$	R	Self Inductance
D.	$\frac{\varepsilon}{\frac{dl}{dt}}$	S	Induced emf

(1) A(P), B(Q), C(R), D(S)

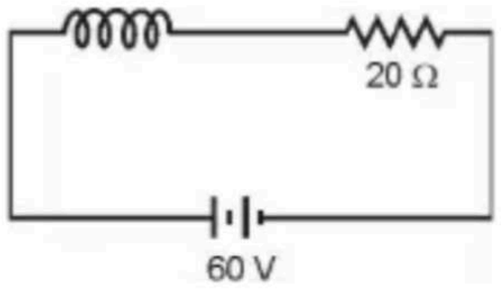
(2) A(S), B(P), C(Q), D(R)

(3) A(S), B(Q), C(P), D(R)

(4) A(Q), B(P), C(S), D(R)

Space for Rough Work

37. Consider the following *LR* circuit, what is the current in the circuit at steady state if inductive coil has inductance of 10 mH, while it offers a resistance of 10 Ω as well?

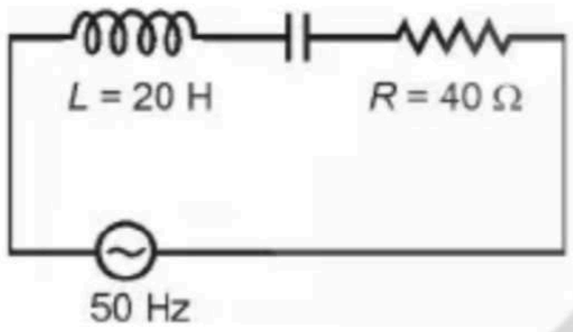


(1) 1 A (2) 2 A
(3) 3 A (4) 4 A

38. Indian electricity grid system works on a natural frequency of 50 Hz. In an *LCR* circuit if inductance is $\frac{20}{\pi}$ mH then for driving it to resonance its capacitive reactance should be

(1) $\frac{2}{\pi}\Omega$ (2) 20 Ω
(3) $\frac{20}{\pi}\Omega$ (4) 2 Ω

39. The average power dissipated in the given circuit connected to an AC main of 50 Hz and carrying current of 4 A, is



(1) 160 W (2) 240 W
(3) 400 W (4) 640 W

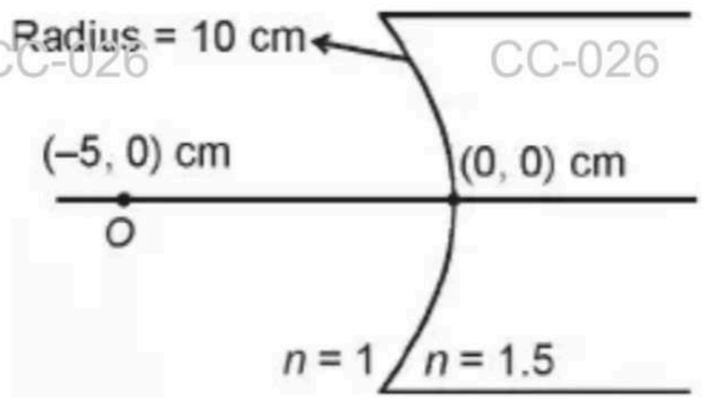
40. A plane electromagnetic wave having intensity 8×10^3 W/m^2 strikes a surface and is completely reflected in the same medium. If refractive index of that medium is 1.5 then radiation pressure exerted by the electro magnetic wave is

(1) 5.33×10^{-5} Pa (2) 2.66×10^{-5} Pa
(3) 4×10^{-5} Pa (4) 8×10^{-5} Pa

41. For a triangular prism with angle of prism is 60° and refractive index 1.5, the angle of minimum deviation will be

(1) $2\sin^{-1}\left(\frac{3}{4}\right)$ (2) $2\sin^{-1}\left(\frac{3}{4}\right)-\frac{\pi}{3}$
(3) $\frac{\pi}{6}$ (4) $\sin^{-1}\left(\frac{3}{2}\right)-\frac{\pi}{6}$

42. An object *O* is placed infront of a refracting surface having radius 10 cm and refractive index 1.5 as shown in figure the position of image for object *O* will be



(1) (4, 0) cm (2) (8, 0) cm
(3) (–4, 0) cm (4) (–6, 0) cm

43. **Assertion (A):** Interference obeys the law of conservation of energy

Reason (R): The energy is redistributed in case of interference.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is not correct
(4) (A) is not correct but (R) is correct

44. Two polaroid's are oriented with their planes perpendicular to incident light and transmission axis making an angle of 30° with each other. What fraction of incident unpolarized light is transmitted?

(1) 20% (2) 45%
(3) 37.5% (4) 75%

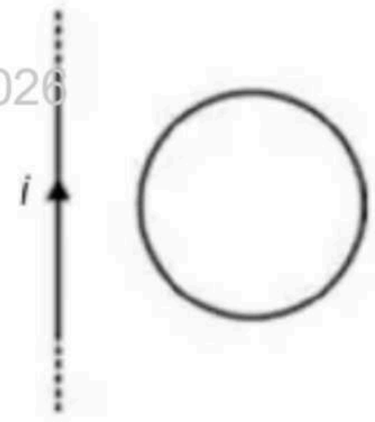
Space for Rough Work

45. A coil has 2000 turns and area of 70 cm^2. The magnetic field perpendicular to the plane of the coil is 0.3 Wb/m^2 and the coil takes 0.1 second to rotate through 180°. The value of the average induced emf will be

(1) 4.2 V (2) 8.4 V

(3) 42 V (4) 84 V

46. A current (i) carrying long wire and a conducting loop are in same plane as shown. If the current (i) in wire is increasing, then induced current in the loop is



(1) Anticlockwise

(2) Clockwise

(3) Zero

(4) First clockwise then anticlockwise

47. The effective value of current $i = 2 \sin 100\pi t + 3 \sin(100\pi t + 60°)$ is

(1) 5 A (2) $\sqrt{13}$ A

(3) $\sqrt{\frac{19}{2}}$ A (4) $\sqrt{\frac{17}{2}}$ A

48. The dimension of $\varepsilon_0 \frac{d\phi_E}{dt}$ is

(1) [AT] (2) [A]

(3) [ML^{2}T^{-2}] (4) [ML^{-1}T^{-2}]

49. A convex mirror has focal length 50 cm in air. If it is submerged in water $\left(\mu = \frac{4}{3}\right)$ then its focal length now will be

(1) 45 cm (2) 50 cm

(3) 80 cm (4) 120 cm

50. S_1 and S_2 are the two coherent point sources of light located in the x-y plane at point (0, 0), and (0, 6λ) respectively. Here λ is the wavelength of light. At which one of the following point (given as coordinates), the intensity of interference will be maximum?

(1) (4λ, 0) (2) (8λ, 0)

(3) $\left(\frac{8\lambda}{3}, 0\right)$ (4) (2λ, 0)

CHEMISTRY

SECTION-A

51. Incorrect statement about nitric acid is

(1) It is formed by catalytic oxidation of NH_3 by atmospheric oxygen

(2) Laboratory grade nitric acid contains ~68% of HNO_3 by mass

(3) Nitric acid on reaction with gold produces NO as one of the product

(4) Aluminium does not dissolve in conc. HNO_3 due to formation of passive film of oxide on the surface

52. Ion present in brown ring complex is

(1) $[Fe(H_2O)_6]^{3+}$ (2) $[Fe(H_2O)_5(NO)]^{3+}$

(3) $[Fe(H_2O)_5NO]^{2+}$ (4) $[Fe(H_2O)_5(SO_4)]^{+}$

53. Oxoacid of phosphorous which contains two 'P – OH' bonds is

(1) H_3PO_2 (2) H_3PO_3

(3) $H_4P_2O_7$ (4) H_3PO_4

54. Disproportionation of Se_2Cl_2 gives

(1) Se and $SeCl_2$ (2) $SeCl_4$ and $SeCl_2$

(3) $SeCl_4$ and Se (4) Se and $SeCl_6$

Space for Rough Work

55. Select the acidic oxide among the following

(1) CrO_3 (2) CrO

(3) NO (4) N_2O

56. Consider the following statements

(a) Rhombic sulphur is readily soluble in CS_2

(b) Rhombic sulphur is also called as β-sulphur

(c) α-sulphur is stable below 369 K and transforms into β-sulphur above this temperature

The correct statements are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (c) only (4) (a), (b) and (c)

57. Which atomic or physical properties of halogens regularly decreases with increase in atomic number?

(1) Covalent radius

(2) Boiling point

(3) Melting point

(4) Standard reduction potential

58. Correct order of oxidizing power of dihalogens is

(1) $F_2 > Cl_2 > Br_2 > I_2$ (2) $F_2 > I_2 > Br_2 > Cl_2$

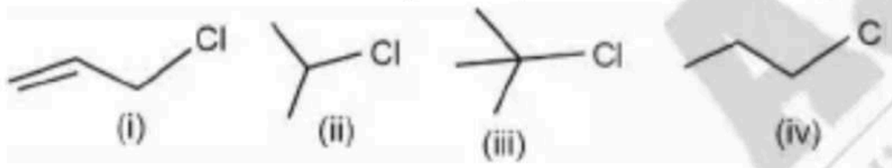
(3) $F_2 > I_2 > Cl_2 > Br_2$ (4) $I_2 > Br_2 > Cl_2 > F_2$

59. Oxyacid of iodine which is least likely to exist

(1) HOI (2) HOIO

(3) $HOIO_2$ (4) $HOIO_3$

60. Consider the following compounds



The correct order of rate of solvolysis of the given compounds in 50% aqueous ethanol in

(1) (i) > (iii) > (ii) > (iv)

(2) (iii) > (i) > (ii) > (iv)

(3) (iii) > (i) > (iv) > (ii)

(4) (i) > (iii) > (iv) > (ii)

61. Consider the following statements

(a) Many copper (I) compounds undergo disproportionation reaction in aqueous medium

(b) E° of Cr^{3+}/Cr^{2+} Couple has a positive value

(c) Density of iron is greater than chromium

The correct statements are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (c) only (4) (a), (b) and (c)

62. Colourless ion among the following is

(1) Ni^{2+} (2) Ti^{4+}

(3) Cu^{2+} (4) V^{3+}

63. Incorrect match among the following is

(1) Mn_2O_7: covalent green oil

(2) V_2O_5: with alkalies gives VO_4^{3-} ion

(3) V_2O_4: with acids gives VO_2^{-} ion

(4) Na_2CrO_4: Yellow solution on acidification turns into orange colour

64. Most basic lanthanoid hydroxide among the following is

(1) $Eu(OH)_3$ (2) $Gd(OH)_3$

(3) $Er(OH)_3$ (4) $Yb(OH)_3$

65. Correct electronic configuration of Am (Americium, atomic no. 95) is

(1) [Rn] $6d^1\ 7s^2$ (2) [Rn] $6d^3$

(3) [Rn] $5f^7\ 7s^2$ (4) [Rn] $5f^9\ 7s^0$

66. Correct IUPAC name of $[Ni(en)_3]Cl_2$ is

(1) Triethylenediaminenickel(II) chloride

(2) Triethylenediaminenickel(II) chlorido

(3) Tris(ethane-1,2-diamine)nickel(II) dichlorido

(4) Tris(ethane-1,2-diamine)nickel(II) chloride

Space for Rough Work

67. Which complex is diamagnetic in nature?

(1) $[Ti(H_2O)_6]Cl_3$ (2) $[Cu(H_2O)_6]Cl_2$

(3) $[Co(H_2O)_6]Cl_3$ (4) $[Mn(H_2O)_6]Cl_2$

68. Crystal field stabilisation energy of complex $[Mn(H_2O)_6]Cl_2$ is

(1) Zero (2) $-1.2\Delta_0$

(3) $-1.6\Delta_0$ (4) $-0.4\Delta_0$

69. Correct electronic configuration of central metal ion in $[Ni(NH_3)_6]Cl_2$ is

(1) $t_{2g}^6 e_g^2$ (2) $t_{2g}^5 e_g^3$

(3) $t_2^6 e^2$ (4) $t_2^5 e^3$

70. An excess of $AgNO_3$ is added to 100 mL of a 0.02 M solution of dichloridotetraaquachromium (III) chloride. The number of moles of AgCl precipitated would be

(1) 0.004 mol (2) 0.002 mol

(3) 0.0002 mol (4) 0.02 mol

71. Given below are the two statements

Statement-I: $(Ph_3P)_3RhCl$ is a Wilkinson catalyst.

Statement-II: Wilkinson catalyst is used in the treatment of lead poisoning.

In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

72. Which one of the following exhibit optical isomerism (en-ethylene diamine)?

(1) cis-$[Pt(NH_3)_2Cl_2]$ (2) $[Ni(CO)_4]$

(3) cis-$[Co(en)_2Cl_2]^+$ (4) trans-$[Co(en)_2Cl_2]^+$

73. Match the following

	List-I		List-II
a.	The metal which reveals a maximum numbers of oxidation states	(i)	Scandium
b.	The metal although placed in 4d series but it is not considered as a transition element	(ii)	Manganese
c.	The transition metal which does not exhibit variable oxidation states	(iii)	Cadmium
d.	The d-block element which shows only +2 oxidation state	(iv)	Zinc

Select the correct option.

(1) a(i), b(ii), c(iv), d(iii)

(2) a(iv), b(ii), c(i), d(iii)

(3) a(ii), b(iii), c(i), d(iv)

(4) a(iv), b(i), c(ii), d(iii)

74. Given below are the two statements

Statement I: S_N2 reactions of optically active halides are accompanied by inversion of configuration.

Statement II: In case of optically active alkyl halides, S_N1 reactions are accompanied by racemisation.

In the light of above statements, choose the most appropriate answer from the options given below.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

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Space for Rough Work

75. Alkyl iodide are generally prepared by
(1) Swarts reaction (2) Sandmeyer reaction
(3) Gatterman reaction (4) Finkelstein reaction

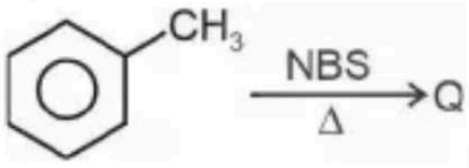
76. Terminal gem dihalides on hydrolysis gives
(1) Ketones (2) Aldehydes
(3) Carboxylic acids (4) Halohydrins

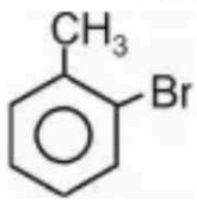
77. Correct order of negative hydration enthalpy of the given ions is
(1) $Co^{2+} > Cu^{2+} > V^{2+} > Fe^{2+}$
(2) $Cu^{2+} > Co^{2+} > Fe^{2+} > V^{2+}$
(3) $V^{2+} > Fe^{2+} > Co^{2+} > Cu^{2+}$
(4) $Cu^{2+} > Co^{2+} > V^{2+} > Fe^{2+}$

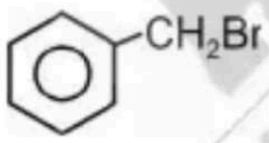
78. Chloroform is slowly oxidised by air in the presence of light to an extremely poisonous gas A. A is
(1) CCl_4 (2) $COCl_2$
(3) CH_3CHCl_2 (4) CCl_3CHO

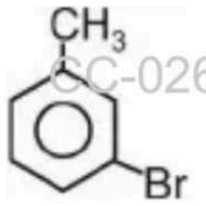
79. Which among the following is an optically active molecule?
(1) $CH_2 = C = C = CH - Br$
(2) $ClCH = C = CHCl$
(3) $Cl_2C = C = CHCl$
(4) $Cl_2C = C = C = CCl_2$

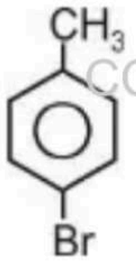
80. The major product (Q) in the following reaction is



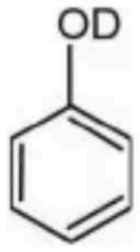
(1) 

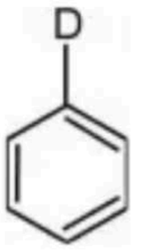
(2) 

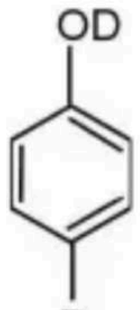
(3) 

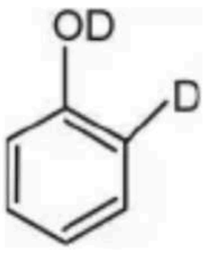
(4) 

81. Phenylmagnesium bromide on reaction with D_2O, gives

(1) 

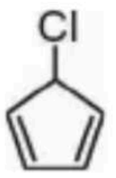
(2) 

(3) 

(4) 

82. Which among the following will react fastest with $AgNO_3$?

(1) 

(2) 

(3) 

(4) 

83. Most suitable reagent for conversion of alcohol into corresponding alkyl halide is
(1) $SOCl_2$ (2) PCl_3
(3) PCl_5 (4) $HCl/ZnCl_2$

84. Consider the following statements
(a) Cyanocobalamine (vitamin B_{12}) is a coordination compound of cobalt
(b) $[CoF_6]^{3-}$ is a spin free complex
(c) In $[Mn(CN)_6]^{3-}$ the central metal ion is sp^3d^2 hybridised.

The correct statement are
(1) (a) and (b) only
(2) (a) and (c) only
(3) (b) and (c) only
(4) (a), (b) and (c)

Space for Rough Work

85. Match the ions in List-I with their calculated magnetic moment value in List-II.

	List-I		List-II
(a)	Sc^{3+}	(i)	5.92 BM
(b)	Ti^{2+}	(ii)	0 BM
(c)	Mn^{2+}	(iii)	2.83 BM
(d)	Cr^{2+}	(iv)	4.90 BM

Choose the correct answer from the options given below.

(1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(2) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(3) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

SECTION-B

86. The geometry and magnetic behaviour of the complex $[Ni(CO)_4]$ are

(1) Square planar and diamagnetic

(2) Tetrahedral and diamagnetic

(3) Tetrahedral and paramagnetic

(4) Square planar and paramagnetic

87. Copper on reaction with dilute nitric acid produces which gas?

(1) N_2 (2) N_2O

(3) NO_2 (4) NO

88. Match the underlined atom of compounds given in List-I with their oxidation state in given List-II.

	List-I		List-II
(a)	$H_4\underline{P_2}O_5$	(i)	+1
(b)	$H_4\underline{P_2}O_6$	(ii)	+3
(c)	$H_3\underline{P}O_2$	(iii)	+4
(d)	$H_4\underline{P_2}O_7$	(iv)	+5

Choose the correct answer from the options given below.

(1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(2) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(3) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

89. For which of the following pairs standard heat of formation is positive?

(1) H_2O and H_2S (2) H_2O and H_2Se

(3) H_2Se and H_2Te (4) H_2S and H_2Se

90. Bronze is an alloy of

(1) Cu and Zn (2) Cu and W

(3) Cu and Sn (4) Cu and Ni

91. Which of the following is an outer orbital complex?

(1) $[Ni(NH_3)_6]^{2+}$ (2) $[Fe(CN)_6]^{4-}$

(3) $[Co(NH_3)_6]^{3+}$ (4) $[Mn(CN)_6]^{4-}$

92. Given below are the two statements, one is labelled as Assertion (A) and other is labelled as Reason (R)

Assertion (A): H_2S is more acidic in nature than H_2O

Reason (R): Bond dissociation enthalpy of H_2S is greater than H_2O.

In light of the above statements, choose the correct answer

(1) (A) is correct but (R) is incorrect

(2) (A) is incorrect but (R) is correct

(3) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

93. The correct order of ligand field strength of the given ligands is

(1) $H_2O < Cl^- <$ en $<$CO

(2) $Cl^- < H_2O <$en $<$CO

(3) $H_2O <$en $<$CO $< Cl^-$

(4) $Cl^- <$ en $< H_2O <$CO

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Space for Rough Work

94. Consider the following statements

(i) Copper does not liberate hydrogen on reaction with dilute H_2SO_4

(ii) Metallic radius of titanium is greater than vanadium

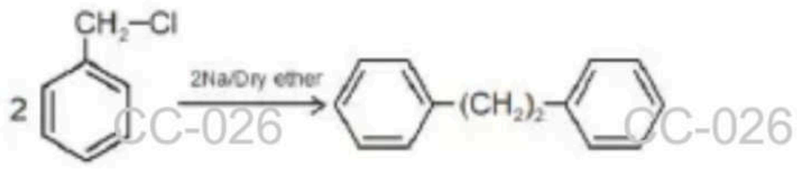
(iii) $Ni(CO)_4$ is an organometallic compound

The correct statements are

(1) (i) and (ii) only (2) (ii) and (iii) only

(3) (i) and (iii) only (4) (i), (ii) and (iii)

95. The given reaction is an example of



(1) Kolbe's reaction

(2) Wurtz-Fittig reaction

(3) Sandmeyer's Reaction

(4) Wurtz reaction

96. Consider the following reaction



The reaction will be fastest in which solvent?

(1) Acetone

(2) DMSO

(3) Ethanol

(4) N,N-Dimethylformamide

97. Total number of optically active isomers of 2, 3-Dibromopentane is

(1) 2 (2) 4

(3) 1 (4) 6

98. Correct order of boiling point of the given compounds is

(1) HF >HI >HBr >HCl

(2) HF >HCl >HI >HBr

(3) HI >HF >HBr >HCl

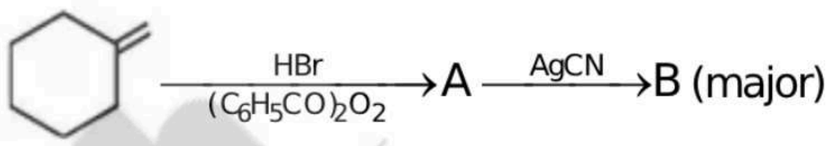
(4) HF >HCl >HBr >HI

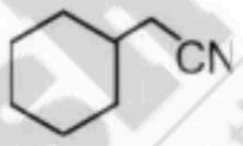
99. The total number of isomers that can be obtained (including stereoisomers) on dibromination of propane is

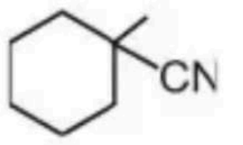
(1) 8 (2) 4

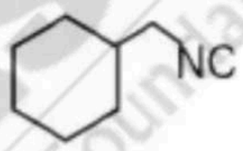
(3) 6 (4) 5

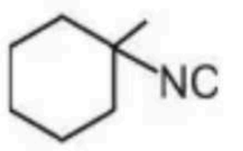
100. The compound (B) in the given reaction sequence is



(1) 

(2) 

(3) 

(4) 

BOTANY

SECTION-A

101. The last step of DNA fingerprinting is

(1) Electrophoresis

(2) Hybridisation using probe

(3) Autoradiography

(4) Blotting

102. Consider the following statements and choose the option which includes **correct** ones.

A. Cytosine and thymine are common in both DNA and RNA.

B. Lambda phage has 5386 nucleotides

C. Along with nucleus, DNA is also found in mitochondria and chloroplast.

D. The backbone of polynucleotide chain is formed due to sugar and phosphates.

(1) A and B (2) C and D

(3) A and C (4) B and D

Space for Rough Work

103. Which among the following enzyme facilitates the opening of DNA helix during transcription?
(1) DNA helicase (2) DNA ligase
(3) RNA polymerase (4) DNA polymerase

104. Who gave the term 'Nuclein' for the acidic substance present in nucleus?
(1) T.H. Morgan (2) Maurice Wilkins
(3) Rosalind Franklin (4) Friedrich Meischer

105. During transcription in prokaryotes, the core enzyme of RNA polymerase is capable of catalysing the process of
(a) Initiation
(b) Elongation
(c) Termination
Choose the **correct** one(s).
(1) (b) only (2) (a) and (b) only
(3) (a) and (c) only (4) (c) only

106. Read the following statements and choose the **correct** option.
Statement A: Genetic code is degenerate and non-overlapping.
Statement B: Marshall Nirenberg's cell free system for protein synthesis helped the genetic code to be deciphered.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

107. Histones are
(1) Responsible for DNA packaging in prokaryotes
(2) Negatively charged proteins
(3) Rich in basic amino acids residues
(4) Synthesized in nucleoid region

108. Select the **correct** statement among the following.
(1) In capping, an unusual nucleotide is added to 3′ end of hnRNA
(2) In prokaryotes, the transcription and translation can be coupled
(3) Prokaryotic cells have only three types of RNA polymerases
(4) During tailing of hnRNA, adenylated residues are added at 3′ end in template dependent manner

109. The genes in eukaryotes are said to be split as they have
(1) Non-functional exons only
(2) Mature mRNA containing both exons and introns
(3) Interrupted coding sequences
(4) Functional introns only

110. Match the column I with column II and choose the **correct** option.

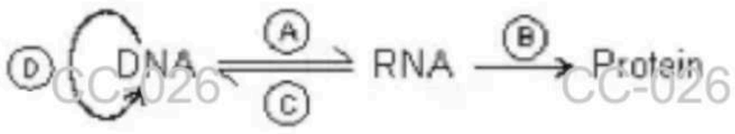
	Column I		**Column II**
a.	*lac y* gene	(i)	It encodes transacetylase.
b.	*lac z* gene	(ii)	It codes enzyme which is primarily responsible for hydrolysis of disaccharide (lactose)
c.	*lac i* gene	(iii)	It codes for an enzyme that increases permeability of the cell to β-galactosides
d.	*lac a* gene	(iv)	It codes for repressor

(1) a(ii), b(iv), c(i), d(iii)
(2) a(iii), b(i), c(iv), d(ii)
(3) a(iii), b(iv), c(i), d(ii)
(4) a(iii), b(ii), c(iv), d(i)

Space for Rough Work

111. Select the **incorrect** statement about RNA polymerase.
(1) It binds to promotor site of DNA segment to start transcription
(2) It uses nucleoside triphosphates as substrates
(3) It polymerises RNA in template dependent fashion
(4) It is involved in elongation of RNA not in the opening of DNA helix during transcription

112. In below flow chart, which step is found in some RNA viruses only?



(1) D (2) A
(3) B (4) C

113. The very fine X-ray diffraction data of DNA was produced by
(1) J. Watson and F. Crick
(2) E. Chargaff
(3) M. Wilkins and R. Franklin
(4) F. Meischer

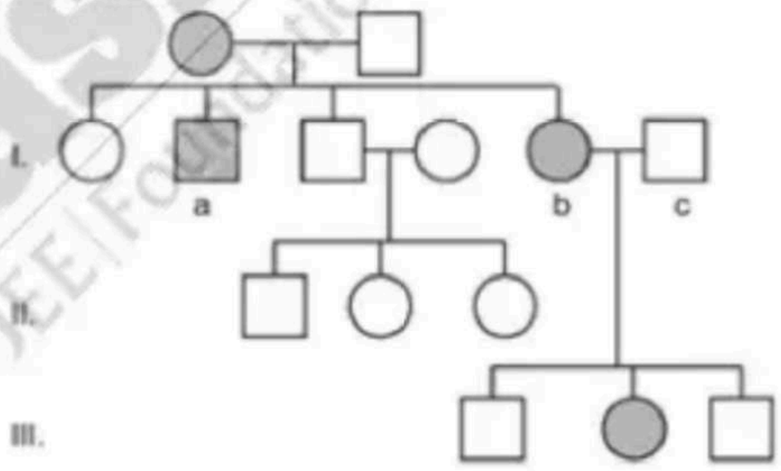
114. Select the **incorrect** statement w.r.t. law of segregation.
(1) The law is based on the fact that two factors of a character do not show any blending
(2) Both the traits are recovered as such in F_2 generation
(3) Gamete receives both the factors of a character during gamete formation
(4) Heterozygous individual produces two kinds of gametes each having one factor with equal proportion

115. Which of the given is not **true** for Down's syndrome?
(1) Occurs due to non-disjunction of two chromosomes of a homologous pair
(2) Affected individual has only one copy of chromosome 21
(3) Affected individual has short stature
(4) It does not follow Mendelian inheritance and thus cannot be shown via pedigree chart

116. In ABO blood grouping
(1) Allele I^A and I^B do not produce any sugar
(2) Allele *i* produces a slightly different form of sugar
(3) A total of four different genotypes are formed
(4) Individuals with AB blood group will never have allele *i*

117. In a clinical diagnosis a female 'A' was identified with absence of secondary sexual characters and has rudimentary ovaries. Which of the given statements can be **true** for the female 'A'?
(1) The chromosome complement of the female cannot be 44 +XO
(2) The female is suffering from a sex linked recessive disorder
(3) Mother of this female may be producing abnormal egg (22 + O) and father producing normal sperm (22 +X)
(4) This female should pass this disorder to her daughters

118. A pedigree chart is given below. Identify **incorrect** option regarding this pedigree.



(1) This pedigree can be applicable for myotonic dystrophy
(2) Above pedigree shows genetic disorder like thalassemia
(3) Male I c can be carrier if the trait is considered for colour-blindness
(4) This pedigree is not applicable for Y linked disorders

Space for Rough Work

119. Mark the **incorrect ratio** w.r.t. genetic crosses.
(1) Monohybrid phenotypic ratio- 3 : 1
(2) Dihybrid genotypic ratio- 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1
(3) Monohybrid genotypic ratio- 1 : 2 : 1
(4) Incomplete dominance phenotypic ratio- 1 : 1 : 1

120. In a family, father is of B blood group and mother is of A blood group. If their children show 50% probability of AB blood group, it indicates that
(1) Both father and mother are homozygous
(2) Both mother and father are heterozygous
(3) Either of the parent is heterozygous
(4) Mother has a genotype $I^O I^O$ and father has genotype $I^A I^B$

121. Read the following Assertion (A) and Reason (R) and choose the **correct** option.
Assertion (A): In sickle cell anaemia, the shape of RBC changes from biconcave disc to elongated sickle like structure.
Reason (R): The mutant haemoglobin molecule undergoes polymerization under low oxygen tension causing the change in shape of RBC.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false statements

122. Mark **true** (T) or **false** (F) for the statements given below and select the **correct** option.
A. Both starch synthesis and seed shape in pea seeds are controlled by single gene
B. Dominance is not an autonomous feature of a gene
C. In *Drosophila* white eye mutation leads to depigmentation in many other parts of the body

	A	B	C
(1)	T	T	T
(2)	F	T	T
(3)	T	F	F
(4)	F	F	T

123. Read the following statements w.r.t. chromosomal aberrations.
a. Alteration of chromosome structure result in chromosomal aberration
b. Loss or gain of DNA segment due to deletion or insertion results in chromosomal aberrations
c. Chromosomal aberrations are commonly observed in cancer cells
Among above statements, choose the **correct** set of statements.
(1) Only a and b (2) Only b and c
(3) Only a and c (4) All a, b, c

124. The phenomenon in which a single gene can exhibit multiple phenotypic expressions is called
(1) Polygenic inheritance
(2) Polymorphism
(3) Pleiotropy
(4) Codominance

125. Colourblindness occurs in about A percent of males and only about B percent of females.
Choose the **correct** option for A and B respectively
(1) 8 ; 4 (2) 4 ; 0.4
(3) 8 ; 0.4 (4) 10 ; 10

126. Which among the following helps in converting milk to curd and also improves its nutritional quality by increasing vitamin B_{12}?
(1) Lactic acid bacteria (2) Yeast
(3) *Streptococcus* (4) *Aspergillus*

Space for Rough Work

127. For the given statements, choose the **correct** option.

Statement-A: Brewer's yeast is used for fermenting malted cereals and fruit juices to produce methanol.

Statement-B: Wine and beer are produced by distillation of the fermented broth.

(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

128. Which statement is **true** for Bt-cotton?

(1) Bt-toxin genes are present in the crop
(2) It is a wild variety of cotton
(3) Useful to get rid of aphids and mosquitoes
(4) Plants are sensitive to insect larvae

129. All of the following features regarding *Nucleopolyhedrovirus* are correct, **except**

(1) Species-specific
(2) Broad spectrum insecticidal application
(3) Have no negative impacts on plants
(4) No negative impacts on non-target insects

130. *Trichoderma*

(1) Is free living fungus
(2) Are effective against several insect pests
(3) Is used in production of statin
(4) Are used to produce biogas

131. Choose the **correctly** matched pair.

	Column-I (Food/Drink)		**Column-II (Microbes used in preparation)**
(1)	Dosa and Idli	–	*Propionibacterium*
(2)	Bread	–	Baker's yeast
(3)	Swiss cheese	–	*Lactobacillus*
(4)	Toddy	–	*Penicillium*

132. Which of the following statements is **wrong**?

(1) Members of the genus *Glomus* form mycorrhiza.
(2) Organic farming is raising of crops through the use of fungi only.
(3) *Rhizobium* fixes atmospheric nitrogen into organic forms, which is used by plants.
(4) In mycorrhiza, the fungal symbiont absorbs phosphorus from the soil and passes it to the plant.

133. Which of the following enzymes(s) is/are used in detergent formulations and helps in removing oily stains from laundry?

a. Pectinases
b. Lipases
c. Proteases
d. Amylase

(1) Only a and d
(2) Only a, b and c
(3) Only b
(4) Only b and c

134. Citric acid is produced commercially by

(1) *Aspergillus niger*
(2) *Clostridium butylicum*
(3) *Acetobacter aceti*
(4) *Methanogens*

135. BOD of waste water is estimated by measuring the amount of

(1) Biodegradable organic matter
(2) Oxygen evolution
(3) Total inorganic matter
(4) CO_2 consumption by microbes

SECTION-B

136. ATGCATTAGCTA is a sequence from the coding strand of a gene. What will be the sequence of the transcribed mRNA?

(1) UAGCUAAUGCAU
(2) TACGTAATCGAT
(3) AUGCAUUAGCUA
(4) TUGCTUUTGCAU

Space for Rough Work

137. Select the **incorrect** match w.r.t. *lac* operon.
(1) Promotor gene – Possess site for RNA polymerase binding
(2) Regulator gene – Functional in the presence of lactose only
(3) Structural gene – Transcribes mRNA for polypeptide synthesis
(4) Operator gene – Interact with a regulatory molecule

138. Through which linkage/bond two nucleotides are linked to form dinucleotide?
(1) Phosphoester linkage
(2) Glycosidic linkage
(3) Hydrogen bond
(4) Phosphodiester linkage

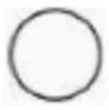
139. Given below are two statements.
Statement I: Severo Ochoa enzyme is polynucleotide phosphorylase.
Statement II: Polynucleotide phosphorylase polymerises RNA with defined sequence in a template independent manner.
In the light of above statements, choose the **correct** answer from the options given below.
(1) Statement I is incorrect but statement II is correct
(2) Both statement I and statement II are correct
(3) Both statement I and statement II are incorrect
(4) Statement I is correct but statement II is incorrect

140. If in a double stranded DNA molecule, the percentage of adenine is 10%, then what will be the percentage of cytosine?
(1) 40% (2) 60%
(3) 20% (4) 10%

141. Which of the following eukaryotic rRNAs is **not** transcribed by RNA polymerase I?
(1) 28S rRNA (2) 5S rRNA
(3) 18S rRNA (4) 5.8S rRNA

142. Linkage
(1) Term was coined by Sutton and Boveri
(2) Describes the generation of nonparental gene combinations
(3) Refers to physical association of genes present on same chromosomes
(4) Is 100% in genes located on different Chromosomes

143. Experimental verification of the chromosomal theory of inheritance was carried by
(1) Sutton and Boveri
(2) T.H. Morgan
(3) Alfred Sturtevant
(4) Henking

144. In pedigree analysis, the symbol for affected male is
(1)  (2) 
(3)  (4) 

145. XO type of sex-determination in which the males have only one X-chromosome besides the autosomes whereas females have a pair of X-chromosomes is shown by
(1) Grasshopper (2) *Drosophila*
(3) Humans (4) Honey bees

146. Which one of the following genotypes can produce only eight different types of gametes?
(1) AaBbcc (2) AaBbccDd
(3) AaBbCcDd (4) AABBccdd

147. Mark the **incorrect** statement w.r.t. IPM.
(1) It is integration of tactics for control of single pest on one or more crops
(2) Maintains situation in which insects are prevented from causing significant damage to crops
(3) It includes use of pest resistant or tolerant varieties
(4) It does not include predators and pathogens

Space for Rough Work

148. Identify the **incorrect** statement w.r.t. antibiotics.
(1) Full potential of penicillin as an effective antibiotic was established by Ernst Chain and Howard Florey
(2) Penicillin was widely used in World War I to treat American soldiers
(3) Alexander Fleming discovered first antibiotic from a fungal source
(4) Antibiotic means against life in the context of disease-causing organisms

149. Read the statements and select the **correct** option.

Assertion (A): DNA fingerprinting is well known for its application in paternity-maternity disputes and criminal identifications.

Reason (R): DNA from every tissue from an individual show the same degree of polymorphism.

In the light of above statements, choose the **correct** option
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

150. Which of the following is a symbiotic nitrogen fixing bacterium?
(1) *Rhizobium* (2) *Azotobacter*
(3) *Azospirillum* (4) *Glomus*

ZOOLOGY

SECTION-A

151. The chorionic villi and uterine tissue become interdigitated with each other and jointly form a structure called 'X'. Choose the incorrect statement w.r.t 'X'.
(1) The 'X' is connected to the embryo through the umbilical cord which helps in the transport of substances to and from the embryo.
(2) It acts as an endocrine tissue and produces several hormones like hCG, hPL, estrogen, progestogens, thyroxine and cortisol.
(3) 'X' facilitates the supply of oxygen and nutrients to the embryo.
(4) It helps in removal of carbon dioxide and excretory/waste materials produced by the embryo.

152. A female went to the gynaecologist after she missed her two consecutive menstrual cycles. After diagnosis, doctor told that this is a case of ectopic pregnancy with her. She asked the doctor more details about it and doctor explained that in ectopic pregnancy there is implantation of
(1) Embryo in the uterus at morula stage
(2) Embryo at the site other than uterus
(3) Defective embryo within the uterus
(4) Embryo in the uterus at blastocyst stage

153. Consider the given statements.
A. There is a gradual decrease in secretions of gonadotropins.
B. There is increase in secretion of hormone produced from corpus luteum.

The above given statements are true for which phase of menstrual cycle in human females?
(1) Follicular phase
(2) Ovulatory phase
(3) Luteal phase
(4) Menstrual phase

154. **Assertion (A):** When estrogen secretion by the follicles begins to rise steeply, the FSH and LH levels increase markedly.

Reason (R): During follicular phase of uterine cycle, there is significant decrease in the secretion of GnRH as well as estrogens.

In the light of above statements, select the correct option.

Space for Rough Work

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

155. Read the given statements and choose the **correct** option.

Statement A: The entry of sperm into the cytoplasm of the ovum induces the completion of the meiotic division of the secondary oocyte.

Statement B: The second meiotic division is unequal and results in the formation of a second polar body and ootid.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is correct

(4) Only statement (B) is correct

156. How many of the feature(s) given below is/are true for 'Saheli'?

a. Developed at CDRI, Lucknow, India

b. Contains similar chemical compound as that present in implants

c. 'Once a week pill'

d. Effective period is much longer than injections

Select the **correct** option.

(1) One (2) Two

(3) Three (4) Four

157. Choose the **odd** one w.r.t. barriers that are reusable and prevent conception.

(1) Vault

(2) Diaphragm

(3) Cervical cap

(4) Condom

158. A person observed fluid discharge and felt slight pain in his genital region. He went to the doctor for checkup and after diagnosis, it was found that he is suffering from a viral STI which cannot be completely cured even if treated properly. Identify the STI and select the **correct** option.

(1) Syphilis (2) Genital warts

(3) Genital herpes (4) Gonorrhoea

159. One could be free of venereal diseases by following all of the given principles, **except**

(1) Avoid sex with unknown partners

(2) In case of doubt, one should go to a qualified doctor for early detection

(3) Always try to use cervical caps during coitus.

(4) Avoid sex with multiple partners

160. Choose the **odd** one w.r.t techniques that involve *in-vitro* fertilization.

(1) ZIFT (2) ICSI

(3) IUT (4) IUI

161. Industrial melanism and Darwin's finches of Galapagos Island are classic examples of the phenomenon named 'P'. Identify 'P' and select the **incorrect** statement w.r.t it.

(1) It is known to affect Hardy-Weinberg equilibrium.

(2) It is a process in which heritable variations enabling better survival are enabled to reproduce and leave greater number of progeny.

(3) It can lead to stabilisation, directional change or disruption.

(4) It is always random and directionless that causes evolution

162. Mammals were descendants of _____ while ancestors of sauropsids were _____. Select the option that correctly fill the blanks respectively.

(1) Thecodonts, Synapsids

(2) Therapsids, Early reptiles

(3) Pelycosaurs, Synapsids

(4) Thecodonts, Early reptiles

Space for Rough Work

163. When heart of fish and reptile were analysed, it was observed that

(1) Both represent homology and indicates common ancestry

(2) Both represent analogy which is a result of divergent evolution

(3) Both represent homology which is a result of convergent evolution

(4) Both represent analogy which is a result of convergent evolution

164. Choose the carnivore dinosaur among the following characterized by bipedal locomotion.

(1) *Triceratops* (2) *Pteranodon*

(3) *Tyrannosaurus* (4) *Brachiosaurus*

165. If, on average, 36% of the loci in a species gene pool are heterozygous, then the average homozygosity of the species should be

(1) 64% (2) 36%

(3) 86% (4) 74%

166. A pregnant woman went to a gynaecologist for her regular checkup. Doctor did ultrasound and observed eyelashes and separation of eyelids in foetus. The pregnant woman went to doctor in/by the

(1) 1st month of pregnancy

(2) End of second trimester

(3) 5th month of pregnancy

(4) End of first trimester

167. Choose the **incorrect** statement.

(1) Mammary glands differentiate during pregnancy and secrete milk after child birth.

(2) The milk produced during the initial few days of lactation is called colostrum which lacks antibodies.

(3) Breast-feeding during the initial period of infant growth is recommended by doctors for bringing up a healthy baby.

(4) Baby suck out the milk through lactiferous ducts which are connected to mammary ampulla.

168. During labor, _____ causes strong uterine contractions, which is produced by the _____ in the human body.

Select the option that **correctly** fill the blanks respectively.

(1) Prolactin, Adenohypophysis

(2) Oxytocin, Hypothalamus

(3) Estrogen, Graafian follicle

(4) Progesterone, Neurohypophysis

169. How many of the structures given below in the box are human male accessory ducts?

Rete testis, vasa efferentia, epididymis, vas deferens, seminiferous tubules, testicular lobules

Select the **correct** option.

(1) Four (2) Five

(3) Six (4) Three

170. Our population was approximately __A__ at the time of our independence which reached close to __B__ mark by 2000 and crossed __C__ in May 2011.

Select the **correct** option to fill the blanks A, B and C respectively.

(1) 350 million, 1 billion and 1.2 billion

(2) 3 million, 7 million and 8 million

(3) 200 million, 600 million and 720 million

(4) 300 million, 700 million and 800 million

171. Given below are certain diseases/defects.

a. Jaundice b. Cleft palate

c. Haemophilia d. SCID

e. Down's syndrome f. Sickle cell anemia

How many of the diseases/defects given above can be detected by amniocentesis?

(1) Six (2) Four

(3) Five (4) Three

Space for Rough Work

172. Read the given statements and select the **correct** option.

Statement A: During post industrialisation period, the tree trunks became dark due to industrial smoke and soots.

Statement B: After industrialization, there was evolution of white-winged moths that were able to camouflage themselves better.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

173. Complete the analogy by choosing the **correct** option.

Tubectomy : Oviduct : : Vasectomy : _____

(1) Vasa efferentia

(2) Vas deferens

(3) Epididymis

(4) Seminal vesicles

174. A tiny finger-like structure which lies at the upper junction of the two labia minora in human females is

(1) Hymen (2) Urethral opening

(3) Clitoris (4) Mons pubis

175. During the course of evolution, the first mammals on Earth were like

(1) Horse (2) Elephant

(3) Shrews (4) Dinosaurs

176. In the test tube baby programme, embryo with upto eight blastomeres is transferred into

(1) Uterus

(2) Fallopian tube

(3) Vagina

(4) Cervical canal

177. Which of the following represents external genitalia of a human female?

(1) Labia majora, labia minora, oviduct

(2) Labia minora, breasts, cervix

(3) Clitoris, labia majora, womb

(4) Labia majora, labia minora, clitoris

178. Organism which was believed to be extinct but was caught in 1938 in South Africa was

(1) *Rana*

(2) Coelacanth

(3) Flying squirrel

(4) Flying phalanger

179. Probable reasons for population explosion is rapid decline in all of the following, **except**

(1) Number of people in reproducible age

(2) Maternal mortality rate

(3) Infant mortality rate

(4) Death rate

180. Combination of which of the following set of hormones can be used by females as implants under the skin for birth control?

(1) Oxytocin and progestogen

(2) Relaxin and estrogen

(3) Estrogen and oxytocin

(4) Progestogen and estrogen

181. Louis Pasteur's view for the origin of life on Earth is that

(1) Life originated within six days

(2) Life originated from pre-existing life

(3) Life originated spontaneously from non-living substances/matter

(4) Life came from other planets

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Space for Rough Work

182. Bryophytes and tracheophytes have possibly evolved from common ancestor(s) named
(1) Chlorophyte ancestors
(2) Progymnosperms
(3) Rhynia-type plants
(4) Psilophyton

183. In human males, spermatogonia are present on the inside wall of
(1) Rete testis
(2) Vasa efferentia
(3) Epididymis
(4) Seminiferous tubules

184. The embryological support for evolution proposed by Ernst Haeckel was disapproved on careful study performed by
(1) Louis Pasteur
(2) Karl Ernst von Baer
(3) S.L. Miller
(4) Hugo de Vries

185. On Galapagos Islands, _____ finches were the ancestral finches from which the other finches evolved.
Choose the option which fills the blank **correctly**.
(1) Insect eating (2) Fruit eating
(3) Cactus eating (4) Seed eating

SECTION-B

186. Select the **odd** one w.r.t. Darwinism.
(1) The fitness, refers ultimately and only to reproductive fitness
(2) Successful organisms have a low rate of reproduction
(3) Struggle for existence
(4) Adaptive ability is inherited and has genetic basis

187. A population will **not** exist in Hardy-Weinberg equilibrium if
(1) Selective mating is present
(2) Recombination will not occur
(3) Gene migration does not take place
(4) There are no mutations

188. Choose the option with **correct** match between Column I and Column II.

	Column I		Column II
a.	Neanderthal man	(i)	Fossils dated back to 15 mya
b.	*Dryopithecus*	(ii)	Probably lived in East African grasslands
c.	*Homo sapiens*	(iii)	Lived in near East and Central Asia between 1,00,000-40,000 years back
d.	*Australopithecus*	(iv)	Arose between 75,000-10,000 years ago

(1) a(iii), b(i), c(iv), d(ii)
(2) a(iii), b(iv), c(i), d(ii)
(3) a(i), b(ii), c(iii), d(iv)
(4) a(ii), b(iii), c(i), d(iv)

189. In human females the three phases of the uterine cycle within uterus in correct order are
(1) Menstrual → Luteal → Ovulatory
(2) Menstrual → Proliferative → Secretory
(3) Follicular → Ovulatory → Luteal
(4) Ovulatory → Luteal → Proliferative

Space for Rough Work

190. How many primary follicles are left in each ovary at puberty in a normal human female?
(1) 60 – 80
(2) 6000 – 8000
(3) 600 – 800
(4) 60,000 – 80,000

191. Fossils are most abundantly found in which type of rocks?
(1) Metamorphic
(2) Igneous
(3) Sedimentary
(4) Volcanic

192. Comprehend the given statements and select the **correct** option.
Statement A: The ovulatory phase is followed by luteal phase.
Statement B: The levels of progesterone exceed the levels of oestrogen in follicular phase.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

193. Choose the **correct** set of haploid cells.
(1) Spermatogonia and spermatozoa
(2) Primary oocytes and spermatids
(3) Secondary oocytes and spermatids
(4) Oogonia and secondary spermatocytes

194. Select the **correct** option w.r.t. middle layer of uterine wall in human females.
(1) Glandular in nature
(2) Undergoes contractions during parturition
(3) Made up of striated, branched muscle fibres
(4) Sheds off due to sudden fall in progesterone levels

195. All of the following are natural methods of contraception, **except**
(1) Periodic abstinence
(2) Withdrawal method
(3) Coitus interruptus
(4) Lippes loop

196. Intra Uterine Device does **not** suppress
(1) Spermatogenesis
(2) Implantation of blastocyst
(3) Fertilizing capacity of sperm
(4) Sperm motility

197. Which of the following is **true** for vasectomy?
(1) Sperm production decreases
(2) Degeneration of testes
(3) Loss of sexual desire
(4) It blocks gamete transport and thereby prevents conception

198. Select the **incorrect** statement.
(1) The most successful story is the evolution of man with language skills and self consciousness.
(2) Study of comparative anatomy and biochemistry provides evidence for evolution.
(3) Earth was supposed to have been formed about 450 million years ago.
(4) Life appeared 500 millions years after the formation of Earth.

199. **Assertion (A):** In humans, testes are situated outside the abdominal cavity within the scrotum.
Reason (R): The scrotum helps in maintaining the low temperature of the testes (2-2.5°C lower than the internal body temperature) necessary for spermatogenesis.
In the light of above statements, select the correct option.

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(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true, (R) is false

(4) Both (A) and (R) are false

200. Complete the analogy w.r.t. convergent evolution.
Lemur : Spotted cuscus : : Anteater :____
Choose the **correct** option.

(1) Marsupial mole

(2) Bobcat

(3) Numbat

(4) Tasmanian tiger cat

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